Marks Scored

4/10/24

Name... Karthika... N..... Class..... Sec.....

Subject... JavaScript..... Test.....

Test.

1) **Variable :-**

A variable is a named memory location used to store data.

Var:

Function - Scoped and can be redeclared and reassigned

let:

Block - Scoped and can be reassigned but not redeclared.

Const:

Block - Scoped and Cannot be reassigned or redeclared

```
Var a = 10;
let b = 20;
Const c = 30;
```

2) **Data types :-**

Data types specify the kind of value stored in a variable.

```
let name = "Karthi"      -> String
let age = 20             -> Number
let Pass = True          -> Boolean
let result               -> Undefined
let value = null         -> Null
```

String -> "Hello"

Number -> 25

Boolean = True (or) false

Undefine : A variable declared without a value.

Null : An intentional empty value.

3) D/b -> ==& === :-

1) == :-

Compares only the value and may convert the data type.

2) === :-

Compares both value & data type.

Example :-

5 == "5"; -> true

5 === "5"; -> false

=== -> is called the strict equality Operator and is generally preferred.

4) Operators:

Arithmetic Operator : perform mathematical Operations.

```
let a = 10          o/p
let b = 3
a+b                 13 -> Addition
a-b                 7 -> Subtraction
a*b                 30 -> Multiplication
a/b                 3.33 -> Division
a%b                 1 -> Remainder
a**b                1000 -> Exponentiation
```

Marks Scored

Name..... Class..... Sec.....
Subject..... Test.....

Assignment Operator assign or update values in variables. Basic operator (=)

let x = 10;

		O/P
x += 5	x = x+5 ,	15
x -= 2	x = x-2 ,	13
x *= 2	x = x*2 ,	26
x /= 2	x = x/2 ,	23

Comparison Operator Compare two values and return True or false

Common Operator : ==, ===, !=, >, <, >= & <=.

	O/P
10>5	T
10<5	F
10=="10"	T
10===10"	F
10 != 5	T
10 >= 10;	T

Logical Operator are used to combine conditions.

&& (AND): True if both conditions are true

|| (OR): True if at least one condition are true

! (NOT): Reverse the result.

```
let age = 20
let hasId = true
```

O/P :-

age >= 18 && hasId : T

age < 18 || hasId : T

!hasId : F

5) D/B Array, Object & Array of Object :-

Array: Stores multiple values in an ordered list. Array elements are accessed using indexes.

```
let fruits = ["Apple", "Banana", "Mango"]
Console.log (fruits[0]);
```

O/P
Apple

Object An object stores data in key-value pairs.

```
let student = {
    name : "karthi"
    age  : 20
};
Console.log (student.name);
```

O/P
Ravi

Array of Object stores multiple objects inside an array.

```
let students = [ { name : "Ramya", age : 20},
                 { name : "karthi", age : 21}]
Console.log (students[0].name);
```

O/P
Ramya

Marks Scored

Name..... Class..... Sec.....

Subject..... Test.....

Array = Multiple ordered values

Object = key - value pairs

Array of Object = Multiple Objects

6) Function:

Function is a reusable block of code that performs a specific task.

```
function great(){
    console.log("Byee");
}
great();
```

O/P

Byee

Parameter :-

A variable listed in the function definition

Argument:

The actual value passed to the function when it is called.

```
function greet(name){
    console.log("Hello" + name);
}
greet("karthi") // name => Parameter
                karthi => karthi => argument
```

Return

The return Statement sends a value back from a function.

```
function add (a, b) {
    return a + b;
}
let result = add (5, 3)
Console . log (result);
```

O/P
= 8

a, b → Parameters
5, 3 → arguments
return a+b → returns the result 8.

7) Even or Odd

Use the modulus Operator (%) to check the remainder after division by 2.

```
let number = 07;
if (number % 2 == == 0) {
    Console . log ("Even");
}
else{
    Console . log ("odd");
}
```

O/P
odd

Marks Scored

Name.....Class.....Sec.....

Subject.....Test.....

8) **for loop**

A for loop can be used to print all even numbers from 1 to 50.

```
for (let i = 1 ; i <= 50; i++) {
    if (i % 2 == 0) {
        console.log(i);
    }
}
```

O/P

2 4 6 8 10 12
14 16 18 20 22
24 28 30 32 34
... 50

9) **Factorial**

The factorial of a number is the product of all positive integers up to that number.

```
let num = 5;
let fact = 1
for (let z = 1; z <= num ; z++) {
    fact = fact * z
}
console.log (num + " ! = " + fact)
```

O/P

5 ! = 120

10) Fibonacci Series

This Series is a Sequence where each number is the Sum of the Previous two numbers.

```
let first = 0;
let Second = 1;
for (let U = 1; U <= 8; U++){
    console.log (first)
Let next = first + Second;
    first = Second
    Second = next
}
```

op
0
1
1
2
3
5
8
13

11) Array :- Even Num

```
let num = [10, 15, 20, 25, 30, 35, 40, 45, 50]
for (let k = 0; k < num.length; k++){
    if (num[k] % 2 == == 0) {
        Console.log (num[k]);
    }
}
```

op
10
20
30
40
50

Marks Scored

Name..... Class..... Sec.....

Subject..... Test.....

12) Array Total

```
let marks = [80, 75, 90, 85, 70]
let total = 0;
for (let U = 0; U < marks.length; U++){
    total = total + marks[U];
}
console.log ("Total = " + total);
```

O/P

Total = 400

13) Object

```
let student = {
    name : "karthi"
    age : 21
    Course : JS
    mark : 85
};
```

O/P

Student Name : karthi
Age : 21
Course : JS
mark : 85

```
Console.log ("Student Name : ", Student.name);
Console.log (" Age : ", Student.age);
Console.log (" Course :" , Student.Course);
Console.log ("Mark :" , Student.mark)
```

14) Array of Objects:

```
let Students = [ { name : "karthi" , marks : 90 }
                 { name : "Maya", marks : 45 }
                 { name : "karthika", marks : 96 }
                 { name : "krishnan", marks : 35 } ];

for (let i = 0 ; i < Students.length ; i++) {
    if Students[i].marks >= 50) {
        Console.log (Students[i].name , Students
            [i].marks);
    }
}
```

o/p

karthi 90

karthika : 96

15) function to find the largest number :-

```
function findLargest (a,b) {
    if (a>b) {
     return a;
    }
    else {
        return b;
    }
}
  Console.log (findLargest (25, 40));
```

o/p

40